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A 2.023 Ga age for the Vredefort impact event and a first report of shock metamorphosed zircons in pseudotachylitic breccias and Granophyre

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Abstract

U–Pb isotope systematics of shock metamorphosed zircon grains from pseudotachylitic breccias and Granophyre from the controversial Vredefort Structure, South Africa, provide new and compelling evidence for an impact origin for this structure. Zircon grains from these rocks exhibit planar microstructures and polycrystalline textures similar to those from the Chicxulub crater breccia, K/T boundary ejecta, and rocks from the Sudbury Structure. A concordant 2023 ± 4 Ma (2σ) age for newly crystallized, unshocked zircon grains from recrystallized pseudotachylitic breccia from the central part of the Vredefort Structure provides a good approximation of the time of impact. This age indicates that the impact post-dates Bushveld magmatism by at least 30 m.y. U–Pb isotopic results for individual, pre-impact zircon grains indicate crystallization ages from about 3060 to 3300 Ma and Pb loss at ca. 2000 Ma. Data for high U grains plot below a discordia line from 3060 to 2023 Ma and indicate both impact- and post-impact related Pb loss. The data and granular morphology of a zircon grain from the Granophyre indicates probable ca. 2.0 Ga and ca. 1.0 Ga Pb loss. Although planar microstructures in zircon are ubiquitous, there are also some unshocked, low-U grains, and these record a ca. 3.1 Ga primary age. The older ca. 3.1–3.3 Ga ages for shocked zircons reflect formation and modification of granitoid crust in the region of the Vredefort Structure prior to and during a metamorphic event at about 3080 Ma.

The resilience of zircon shock features to post-impact alteration and annealing, in combination with precise U–Pb dating of individual shocked grains provide a valuable method for identifying ancient, metamorphosed and tectonically modified impact structures.

Keywords: Vredefort Dome; U/Pb; zircon; shock metamorphism; South Africa; meteor craters; absolute age

1. Introduction

For much of this century, the Vredefort Dome, located approximately 120 km to the southwest of Johannesburg, South Africa, has been the object of scientific controversy between proponents of either an impact origin for this structure [1–3] or an endogenic origin, by either tectonic or so-called cryptoexplosion processes (refer to [4]; for detailed references see [5]). The near-circular Vredefort Dome is about 70 km wide and is located in the central part of the

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economically important Witwatersrand Basin (Fig. 1a). It consists of a core of Archean granitoid gneisses that is surrounded on the western and northern sides by upturned and partially overturned Archean to early Proterozoic supracrustal strata (Fig. 1b).

The proposal that the Vredefort Structure was formed by hypervelocity impact has gained acceptance, due to the presence of the following shock metamorphic features in Vredefort rocks:

1. shatter cones [6];
2. pseudotachylite [7–10];
3. the high-pressure silica polymorphs coesite and stishovite [11–13];
4. planar microdeformation features in quartz.

This latter evidence has been controversial, because the characteristics of these features do not conform with the criteria for bona fide, shock (impact) diagnostic PDFs (planar deformation features; e.g.,

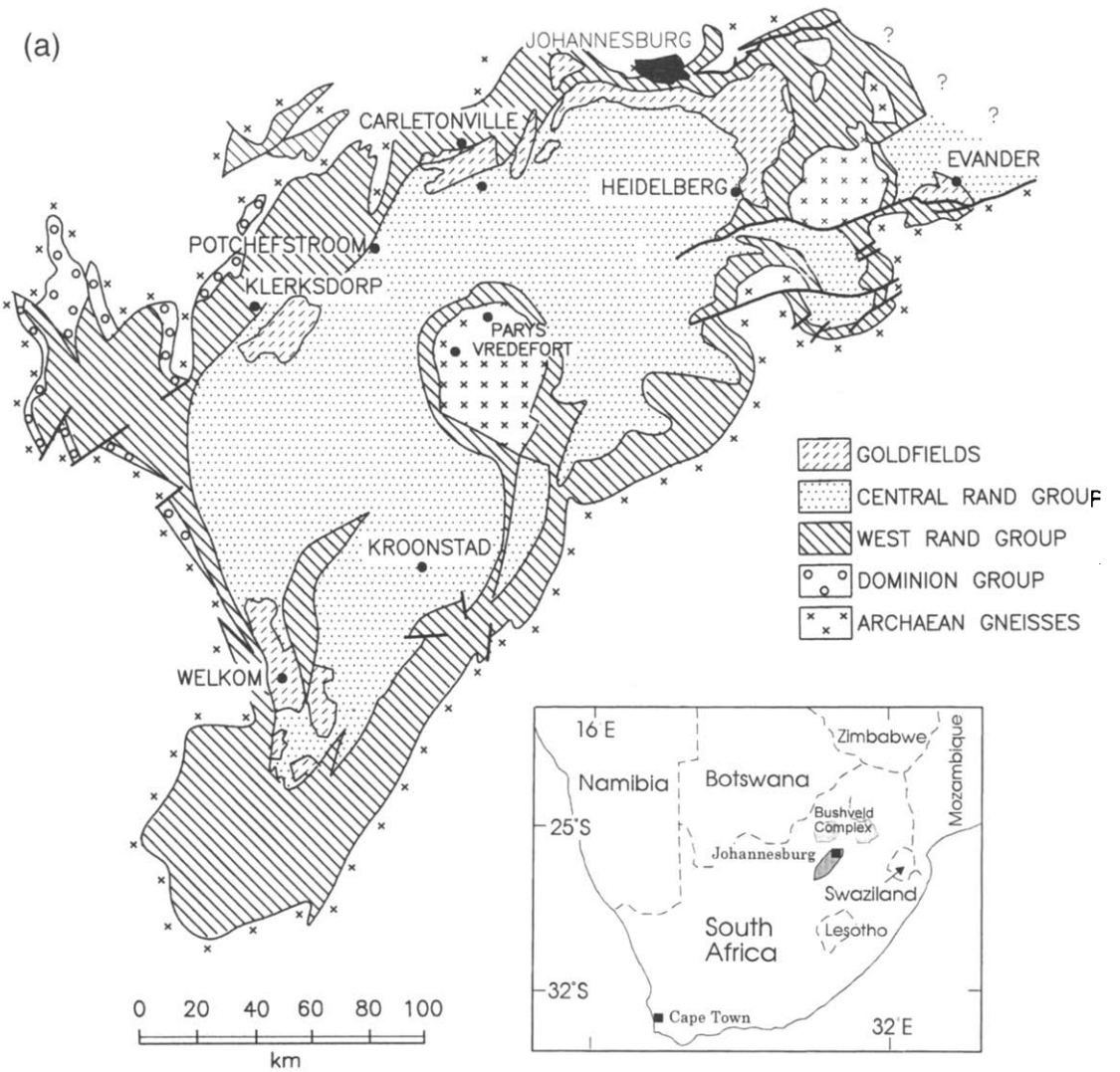


Fig. 1. (a) Position of the Vredefort Dome in the center of the erosional remnant of the Witwatersrand Basin. The inset shows the geographical location of the Witwatersrand basin in southern Africa (dark grey). (b) Schematic geological map of the Vredefort Dome, after [40]. * = sampling locality. *INL* = Inlandsee pan; *LEE* = Leeukop Quarry; *LES* = Lesutaskraal Quarry; *Gr* = Granophyre.

[14,15]). However, a recent Transmission Electron Microscopy study of such features in Vredefort quartz [16] has shown that they represent basal Brazil twins, which have only been described as the result of deformational processes from impactites. The anomalous characteristics of these basal deformation features are the result of post-formational annealing, for which there is widespread evidence in Vredefort rocks.

A suite of dykes of so-called *Granophyre*, a rock type of unique chemical composition and regional homogeneity [17–19], occurs within the granitic basement of the Dome and along the contact between supracrustal collar and basement core. These rocks have long been considered as possible remnants of impact melt but, to date, no shock metamorphosed clasts have ever been identified in this breccia and attempts to identify chemically traces of the

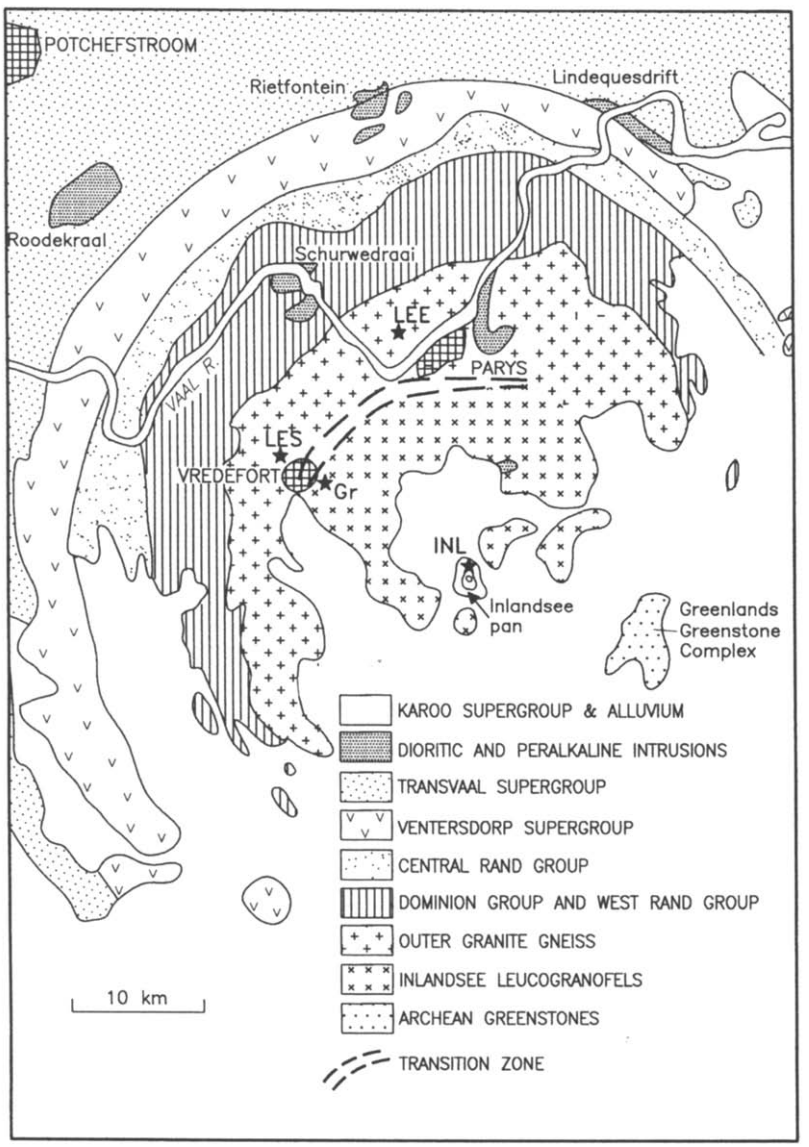


Fig. 1 (continued).

meteoritic component failed [20]; thus, confirmation of its origin as impact melt was not obtained. Only a recent Re–Os isotopic study of Granophyre samples and country rocks [21] indicated a minute contribution of a meteoritic component to this enigmatic rock type. Geochronological results suggest that the Vredefort Granophyre was formed at about 2.0 Ga [22,23], which has been considered the probable time of the impact event.

The Vredefort Dome is the type locality for *pseudotachylite* [7], a breccia characterized by black, aphanitic groundmass that comprises melt-bearing and purely clastic varieties. This breccia type, considered by many structural geologists to be the result of frictional melting [24], has been described, in the form of thin veins rarely exceeding widths of a few tens of centimeters, from numerous tectonic fault zones worldwide. However, the volumes of such breccias in the Vredefort Structure [10] and in the Sudbury impact structure in Ontario, Canada [25], are legendary and in a class of their own. Voluminous occurrences of pseudotachylitic breccias not only occur in the core and collar of the Vredefort Dome, but also in the wider environs, in parts of the Witwatersrand Basin [10]. Results from (whole-rock) $^{40}Ar-^{39}Ar$ step heating dating of such breccias [26] from the northwestern and central parts of the Witwatersrand basin indicate a 2006 ± 34 Ma (2σ) formation age.

The fact that some narrow breccia veinlets in the northeastern part of the Vredefort collar were shown to contain the high-pressure polymorphs coesite and stishovite [11] has long been taken as evidence that all pseudotachylitic breccias in and around the Vredefort Dome were the result of the impact event. Field observations of multiple generations of breccias, in both the Vredefort Dome and the wider Witwatersrand Basin, suggested multiple phases of pseudotachylite formation not necessarily all related to the impact event. This hypothesis was supported by multiple ages obtained by $^{40}Ar-^{39}Ar$ step heating methods on Vredefort pseudotachylite samples [27,28], which suggested pseudotachylitic breccia had been formed not only in the impact event at about 2.0 Ga, but also at later times between 1.4 and 1.1 Ga. However, subsequent laser Ar dating [29] unequivocally showed that the post-2.0 Ga ages were the result of late (ca. 1100–1500 Ma) hydrothermal alteration. Multiple generations of pseudotachylitic breccias have been documented in the Sudbury Structure [30], and have been related to different stages of crater development.

As a result of recent studies by Leroux et al. [16], White [13], and Koeberl et al. [21], residual doubt about an impact origin for the Vredefort Dome has been largely dispelled. Scaling models based on the spatial distribution of deformation features in the Vredefort Dome and wider Witwatersrand basin favour a diameter for the original impact structure (here termed the 'Vredefort Structure') of 200 to > 300 km [31], which is of the same magnitude as the diameters of the other known large impact structures on Earth, the Sudbury [32] and the Chicxulub structures [33,34]. The Vredefort Structure would, thus, have encompassed the whole region known as the Witwatersrand Basin. The Vredefort Dome (Fig. 1b) is considered the deeply eroded remnant of the central uplift of this gigantic impact structure.

Previous, mainly argon chronological, studies have suggested that the Vredefort event occurred ca. 30–40 m.y., after the emplacement of the 2050–2065 Ma Bushveld Complex [22,35]. However, the Ar dating results do not unequivocally define the age for Granophyre emplacement and major pseudotachylitic breccia formation, largely because of post-impact disturbances of the argon isotope systematics. U–Pb dating of zircons has proven to be an effective tool in studies of impact-related sites, because both the time of an impact event and the primary ages of the target rocks can be determined with accuracy and precision. For example, U–Pb zircon dating successfully determined the age of the Sudbury Igneous Complex [36], which is widely regarded as the impact melt body of this large impact structure [32], and underlying basement gneiss, as well as the age of the Manicouagan impact event [37].

The objectives of this study were to apply the U–Pb single grain dating technique to zircon crystals from several Vredefort samples with the hope of determining both a precise age for the time of impact, which is important for distinguishing the effects of the Vredefort event from those of Bushveld magmatism, as well as primary ages for the target rocks. Ages for the latter could provide some new insight into the early history of the basement to the Witwatersrand Basin, and the Kaapvaal craton as a whole.

Pseudotachylitic breccias, Granophyre, and the charnockitic host rock of one breccia sample were collected for dating.

2. Geology of the Vredefort Dome

Numerous detailed geological descriptions of the Vredefort Dome have been published (e.g., [5,38,39]) and for this reason only the main lithological features are indicated in Fig. 1b and discussed here. The Vredefort Dome is conventionally divided into two parts: the central core composed of strongly migmatized Archean granitoid gneisses, and the surrounding collar of largely upturned or overturned, late Archean to early Proterozoic strata belonging to the Dominion Group and the Witwatersrand, Ventersdorp, and Transvaal supergroups [40].

The granitic to granodioritic gneisses of the core are subdivided into the central Inlandsee Leucogranofels (ILG) terrane, and an outer annulus of Outer Granite Gneiss (OGG). The ILG is mainly composed of K-feldspar and quartz-rich gneisses of granulite metamorphic grade, which also host large xenoliths of mafic granulites, high-grade metapelites, and a banded iron formation [41]. The composition of OGG gneisses is generally more heterogeneous than that of the ILG and comprises amphibolite grade tonalitic, granitic, adamellitic, and granodioritic gneisses. Pseudotachylitic breccias are especially prominent in a wide annulus along the contact between these two gneiss types, the so-called Transition Zone [42]. Note that here the term 'pseudotachylitic breccia' is preferred over the term 'pseudotachylite', in reference to its impact-derived origin. This clarification is made because it has been recognized that several breccia types, including clastic and melt breccias (some of which are the result of faulting-related brecciation and/or frictional melting, others formed by shock melting and brecciation), have been collectively referred to as Vredefort pseudotachylite (for a more detailed discussion of this problem, refer to [43]). The Transition Zone is also the region where large amounts of charnockitic (orthopyroxene-bearing) rocks occur. The timing of charnockitization is uncertain: hornblende yielded a 3.03 ± 0.06 Ga (2σ) $^{40}Ar-^{39}Ar$ plateau age [28], whereas a Sm–Nd age of 2.5 Ga on whole rock and mineral separates was interpreted as indicating a metasomatic event [42].

The formation ages for the granitoid gneisses of the core to the Vredefort Dome are poorly constrained. Hart et al.'s [44] early work suggested that the OGG was formed about 3.08 Ga ago. Certain parts of the ILG terrane yielded > 3.5 Ga whole rock Rb–Sr and U–Pb ages, as well as an indication for a possible event about 2.8 Ga ago. Recent $^{40}Ar-^{39}Ar$ step heating results on amphiboles from various core and collar rocks suggest, however, that the 3.08 Ga event is the result of thermal metamorphic overprinting [28,45].

3. Sample descriptions

Five samples from four locations were selected from both ILG and OGG sites and their locations are shown in Fig. 1b. Sample INL was comparatively small, about 700 g, whereas the other four samples were composite samples weighing in total up to 15 kg.

Sample INL is derived from a drillcore obtained at the northern side of the Inlandsee pan just south of the geographical center of the Vredefort Dome. The sample was taken at 139.10 m depth, a few centimeters below the top contact of an about 45 cm wide intersection of pseudotachylitic breccia in ILG type gneiss. The breccia consists of up to several centimeter wide clasts of typical ILG material set into a fine-grained, crystalline, mosaic-textured matrix of euhedral to anhedral quartz, feldspar, biotite and pyroxene (Fig. 2a). Most clasts are completely recrystallized (Fig. 2a) and consist of quartz, K-feldspar, and, locally, partially melted biotite. A general feature of ILG samples from the center of the Dome is the presence of millimeter to, at maximum, 1 cm wide pockets of micropegmatite with granophyric texture, believed to be the result of partial melting. A small body of so-called Central Anatectic Granite [38] at the eastern shore of the Inlandsee pan, a biotite granite believed to represent the product of bulk anatexis of ILG type gneiss, has an imprecise Rb–Sr whole rock age of 1.95 ± 0.05 Ga [44].

The matrix of the INL breccia is relatively 'coarse-grained' (25–50 μm) in comparison with the average grain sizes observed in pseudotachylitic Vre-

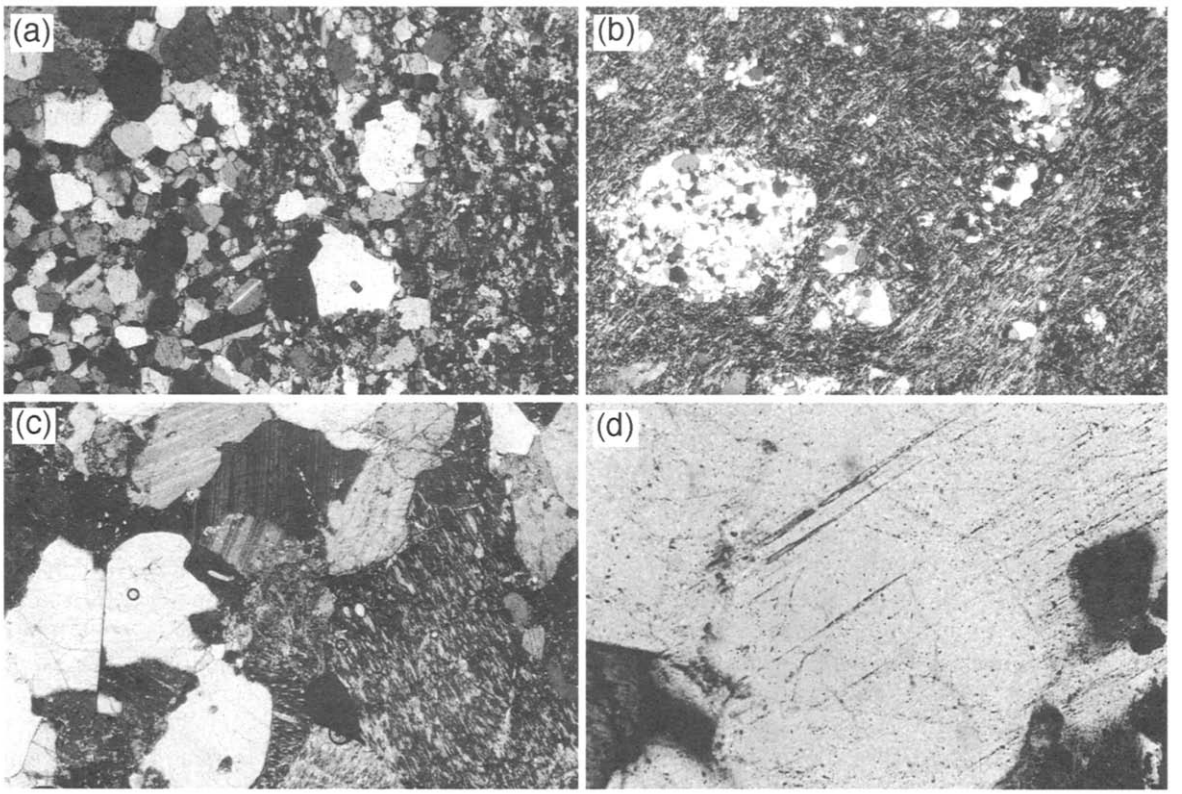


Fig. 2. (a) Typical photomicrograph of pseudotachylitic breccia INL, with recrystallized clasts set in a fine-grained, yet fully crystalline matrix. Width of field of view: 3.4 mm; crossed nicols. (b) Matrix to pseudotachylitic breccia LEE, with flow-aligned plagioclase laths. Several clasts are completely recrystallized and only composed of mosaic quartz. Crossed nicols, 3.4 mm wide. (c) Typical texture of charnockitic gneiss from Lesutaskraal. Note altered pyroxene crystal at lower right. Crossed nicols, 3.4 mm wide. (d) A shocked quartz grain with basal deformation features (according to [16] these features represent basal Brazil twin lamellae) in LES charnockitic gneiss. Crossed nicols, 1.1 mm wide.

defort breccias with crystalline matrices. For this reason this sample was chosen for zircon separation. Zircon crystals of different types were visible at the scale of the optical microscope.

Two samples (denoted LEE and LES in Fig. 1b) were collected within the OGG terrane. Sample LEE is a massive, pseudotachylitic breccia, several metres wide, from the Leeukop Quarry of Marlin Corporation Pty. Ltd., located about 2.5 km due west of the town of Parys. The host rock to the breccia is typical OGG, strongly migmatized and well banded, with alternating granitic and granodioritic bands a few centimeters to tens of centimeters wide. The breccia matrix (Fig. 2b) is largely crystalline, but grain size is much reduced (< 25 μm) compared to sample INL. Cooling below solidus occurred at a rapid rate, judging from the grain size of matrix minerals (mainly plagioclase laths, with fine-grained quartz, amphibole, magnetite, and biotite) and their textures. Clasts in this material range up to several centimeters in size, but all are completely annealed and composed, to a large extent, of relict quartz only (Fig. 2b). Grain shapes are generally well rounded and also indicate melting of the precursor material.

Locality LES (Fig. 1b) refers to a small quarry in predominantly charnockitic gneiss, the precursor material to which can, in places, still be recognized as typical, migmatized OGG in amphibolite facies. LES is located about 1.5 km north of the town of Vredefort on farm Lesutaskraal 604. Like the Leeukop Quarry, the Lesutaskraal exposure represents one of the most prominent occurrences of pseudotachylitic

breccia in the Vredefort Dome. Several metres wide exposures of network breccia occur at a number of sites within this quarry. Matrix zones tens of centimeters wide, surround centimeter to decimeter wide, generally well rounded clasts of charnockitized gneiss. Zircons were separated from composite samples of charnockitic country rock (Ch) and of pseudotachylitic breccia (LES), respectively.

The charnockitic rock is compositionally heterogeneous, with mafic and more felsic bands alternating at a centimeter to decimeter scale. A mafic sample of this material has been described by Reimold et al. [28]. This gneiss generally comprises quartz, plagioclase, K-feldspar, orthopyroxene, amphibole, and biotite, with subordinate magnetite and other heavy minerals. Fig. 2c is a photomicrograph of Lesutaskraal gneiss of intermediate composition, with significant amounts of both felsic and mafic minerals. Orthopyroxene is generally strongly altered, whereas both amphibole and biotite nearly always appear rather fresh. This may indicate that the current mineralogical composition is the result of post-charnockite retrograde metamorphism.

Fig. 2d shows a quartz grain from Lesutaskraal charnockitic gneiss, which displays a prominent set of basal deformation features of the type described by Leroux et al. [16]. Until now, shock metamorphic effects have only been described from quartz in Vredefort rocks. In the course of this investigation we identified in a thin section of the LES charnockitic

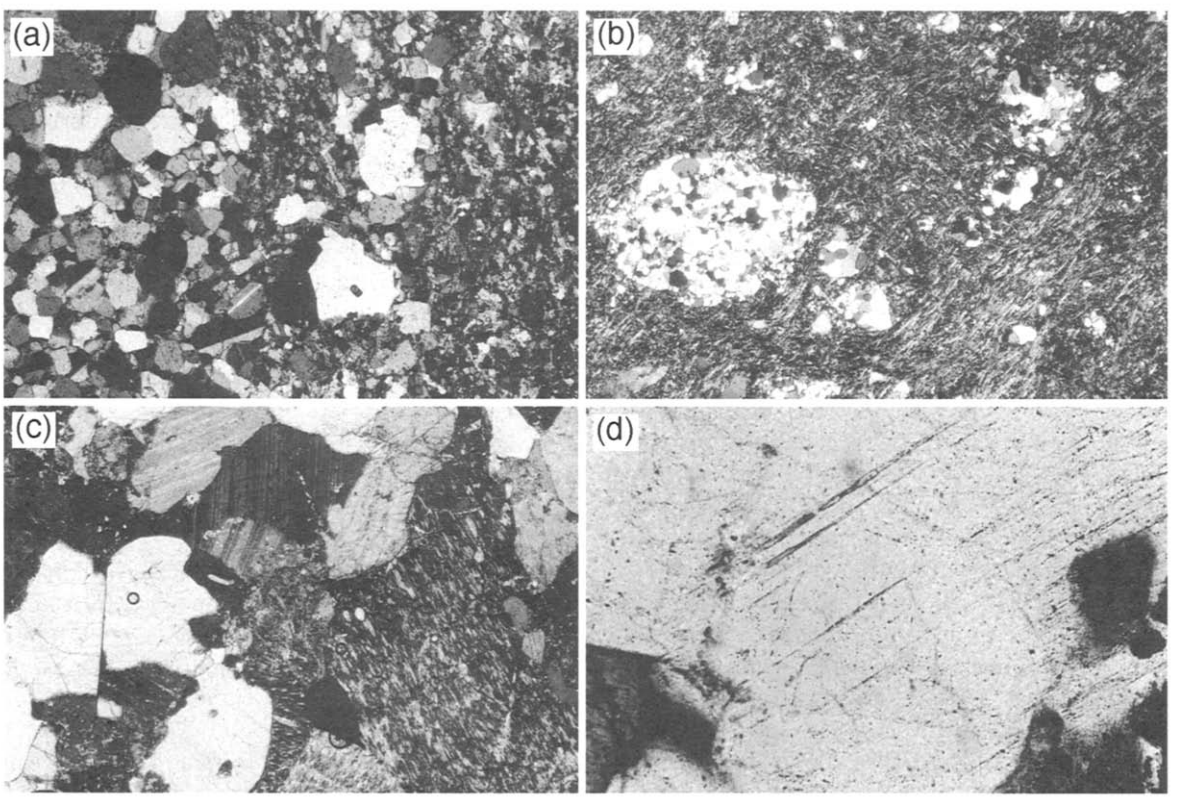
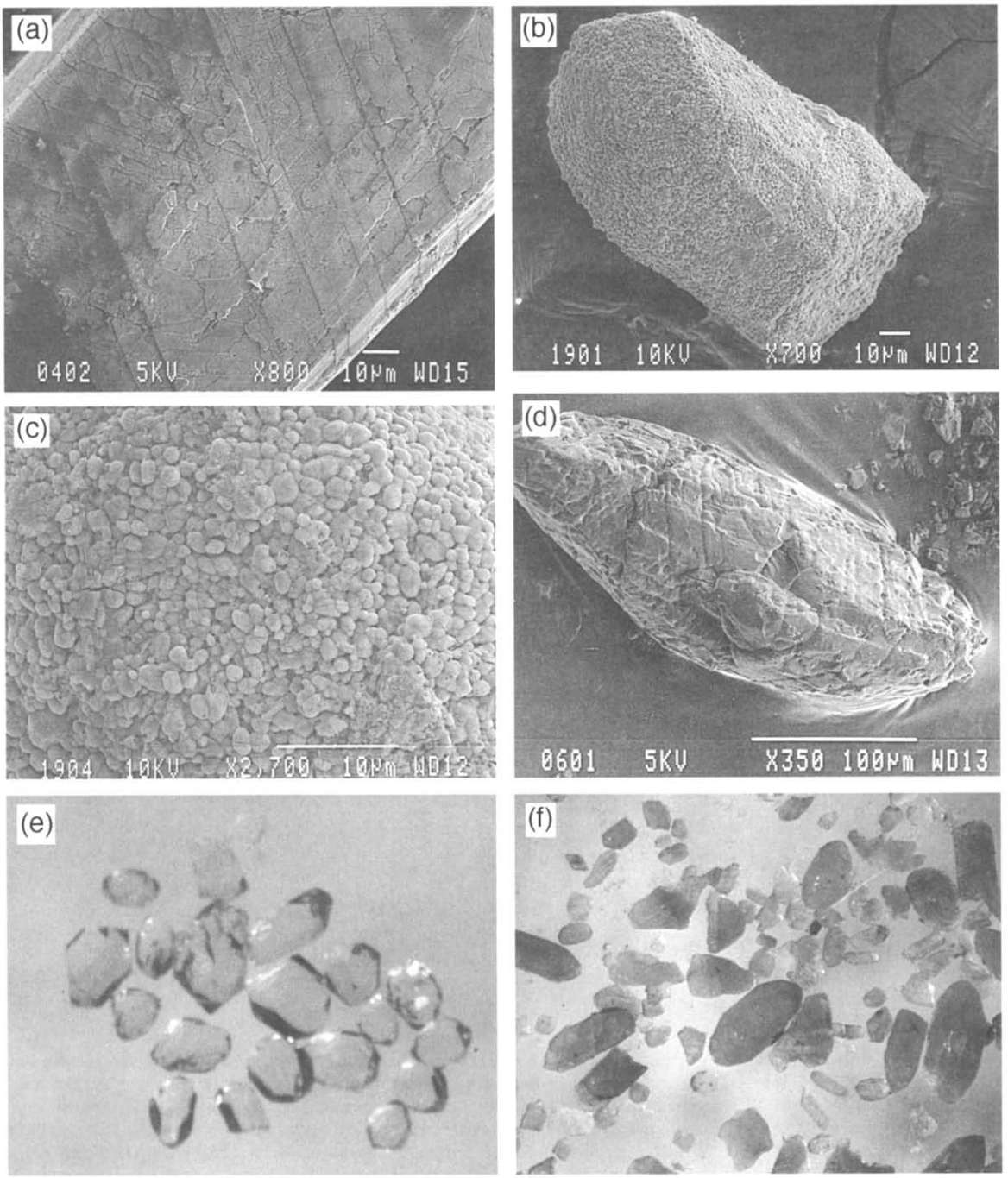


Fig. 3. (a) Well-rounded zircon grain with overgrowth and multiple sets of deformation features from charnockitic gneiss (Ch) from locality LES. Crossed nicols, 355 μm wide. (b) Typical appearance of fine-grained crystalline LES pseudotachylitic breccia, with fully annealed clasts consisting mainly of recrystallized quartz and oxides after thermally decomposed mafic minerals. Crossed nicols, 3.4 mm wide. (c) Fine-grained crystalline matrix of LES pseudotachylitic breccia (note small plagioclase crystals and magnetite crystallites, the latter being responsible for the dark color of these breccias). Crossed nicols, 355 μm wide. (d) Photomicrograph of spherulitic Granophyre. Radial growth of hypersthene needles emanates from a nucleus of fully annealed quartz. Note granular pyroxene crystals in between laths. Plane polarized light, 3.4 mm wide.

gneiss sample a large, well-rounded zircon grain (Fig. 3a) with planar microdeformations. This prompted a detailed analysis of separated zircon grains for possible shock-diagnostic deformation textures.

The textural appearance of LES pseudotachylitic



breccia is very similar to that of the Leeukop breccia (Fig. 3b), except for locally finer matrix in LES material (Fig. 3c). In certain breccia samples from Lesutaskraal, an aphanitic matrix was observed [27,28], whereas thin sections of such samples from this study indicated incipient and, locally, more advanced crystallization. Clasts, again, are nearly always completely annealed and well rounded. The mafic minerals in clasts have been completely converted to oxide assemblages.

A Granophyre (Gr) sample was obtained from a northeast–southwest trending dyke located to the southeast of the town of Vredefort (northeast of farm Holfontein 44). This dyke, described in detail by Therriault [19], consists of both granular and spherulitic (Fig. 3d) melt breccia, with textures vary-

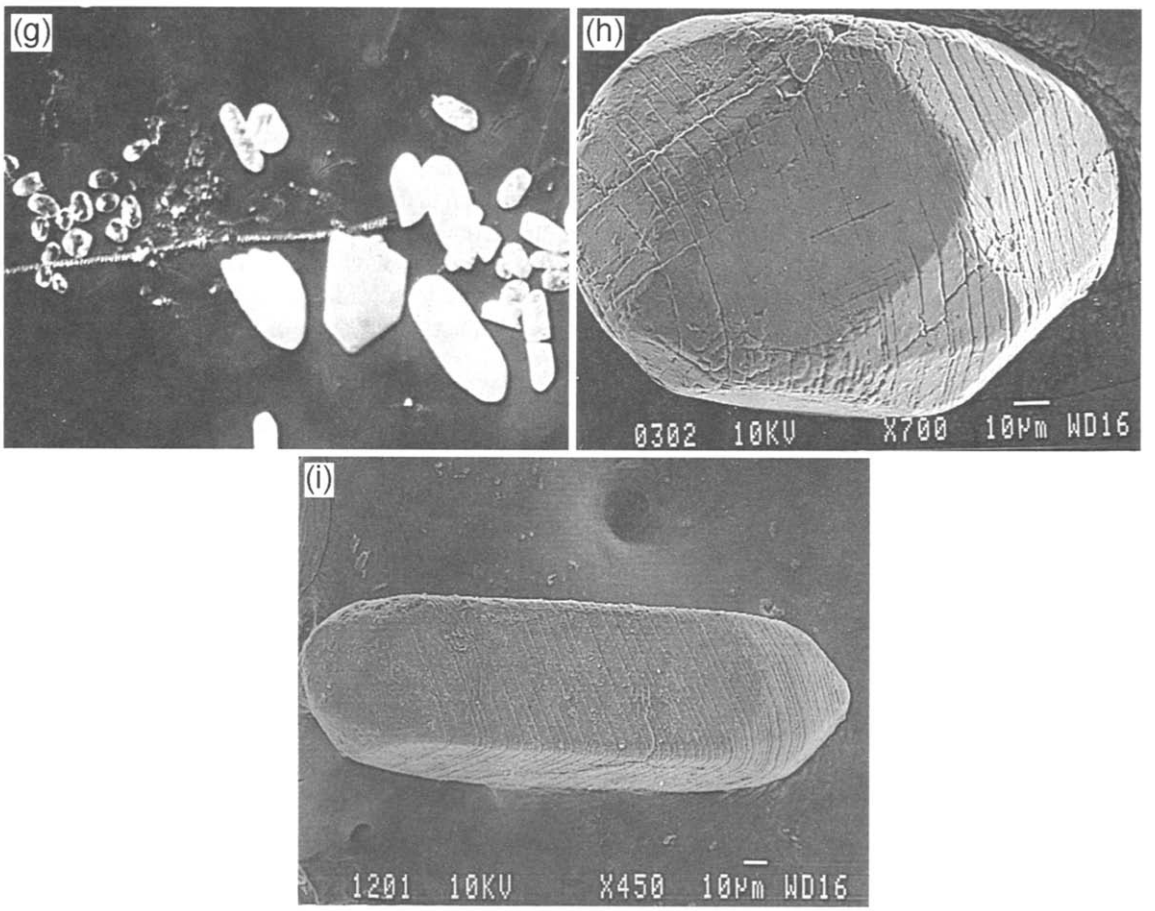


Fig. 4. (a) SEM photograph of a zircon crystal from the Lesutaskraal pseudotachylitic breccia (LES) exhibiting planar microstructures in three directions. (b) and (c) SEM photographs of a granular zircon grain from the Granophyre (Gr) sample. Note the fine degassing holes in some of the granules in (b). This texture has been described previously in [49] and [50] and may be the result of incipient melting at extremely high temperatures (the melting point of zircon is 1676°C). (d) SEM photograph of a typical zircon from the Inlandsee pseudotachylitic breccia sample (INL) showing remnant planar microstructures after possible partial annealing of original shock features. (e) Euhedral, pristine zircons from the Inlandsee pseudotachylitic breccia sample (INL). Width of field of view ~ 1 mm. (f) Representative zircon population from the Lesutaskraal pseudotachyltic breccia sample (LES). Many grains exhibit shock features that are visible using a binocular microscope. Most are opaque to translucent, probably due to alteration and/or scattering of light from shock features. Field of view 2 mm. (g) Photograph of zircons from the charnockite rock sample (Ch) as they appear under a binocular microscope. The small, clear grains on the left side give concordant data, the large, opaque zircons on the right represent the main population and were not analyzed for U and Pb (2 mm field of view). (h) SEM photograph of a shocked zircon crystal from the charnockite rock sample, and (i) from the Granophyre. Note planar microstructures.

ing locally from mainly micropegmatitic intergrowths with variable amounts of needle- or lath-like hypersthene to fully granoblastic matrix with crystal sizes up to > 150 μm. As shown by previous workers [18–20], the overwhelming majority of clasts are completely annealed, but their origin from mainly granitoid precursor rocks is often recognizable. No unequivocal shock deformation effects have been described to date from clasts in Vredefort Granophyre, despite the fact that numerous non-annealed clasts have been studied by many different workers.

4. Shock metamorphosed zircons and their U–Pb systematics

Shock metamorphic features in zircon grains have been reported in breccia from the Chicxulub Structure [46] and associated K/T boundary ejecta from several sites [46–49], and in rocks from the Sudbury [36,49,50], Manicouagan [49], and Haughton impact structures [51]. Planar microstructures are generally resistant to post-impact annealing [36,46–50] and are observed as single or multiple sets of parallel (Fig. 4a), narrowly spaced (≤ 5 μm), planar features that are continuous across the grain. In addition to the planar microdeformation phenomenon, zircon has been found to record another distinct type of shock effect, called the *granular texture* [49] (Fig. 4b,c). This texture is produced by partial or complete recrystallization of a zircon crystal into about 1 μm composite granules and has been observed in laboratory experiments [52]. The presence of granular zircon has been documented in K/T boundary ejecta and Chicxulub impact breccia [46–48], the Onaping Formation in the Sudbury Structure [50], which is interpreted to represent suevitic impact breccia, and in a Muong–Nong tektite [49]. However, zircon of this granular texture has never been described from endogenically deformed terrestrial rocks [49] and, thus, also provides a diagnostic instrument for the identification of impactites. In rocks that contain few quartz crystals, such as in extraterrestrial rocks, the identification of shocked zircons represents a potentially useful tool.

U–Pb dating of individual shocked zircons has proven to be an effective method for deciphering the history of presumed impact-related rocks [46–48,50]. Zircon is a robust, refractory mineral that is morphologically and isotopically resistant to post-impact thermal overprinting. The coupled U–Pb decay scheme in zircon offers the inherent advantage of recording both the original age of the target rocks and the time of isotopic resetting. The primary age is best determined from zircons that exhibit weak shock effects, such as planar deformation features, or no visible shock effects at all. Zircon grains that have been extensively shocked and display a granular texture are subject to the greatest amount of Pb loss and can be useful in determining the time of an impact event. This correlation between the extent of physical shock metamorphic effects in the zircon crystals and the degree of U–Pb fractionation can be considered diagnostic behaviour under the extreme conditions caused by an impact. U–Pb dating of zircons that have crystallized in an impact melt provides the most favourable means by which to date an impact event. This was successfully accomplished for the Manicouagan impact [37], and impact melt lithologies of the Sudbury Structure [36,53,54].

5. Analytical methods

Most zircon grains that were analyzed for U and Pb in this study were first etched in a hot, saturated (90°C) NaOH solution in order to enhance the visibility of planar features [49], and then observed using a scanning electron microscope to document shock features. Selected zircon populations and individual grains from this study are shown in Fig. 4.

Zircon samples selected for isotopic analysis were dissolved, and U and Pb were chemically isolated by using anion exchange columns, following the procedures described by Krogh [55] but using smaller columns (0.05 ml of resin). Procedural blanks were approximately 2.0 pg and 0.5 pg for Pb and U, respectively. Initial common Pb in excess of blank was corrected for using the model of Stacey and Kramers [56]. Isotopic ratios were measured on a VG354 mass spectrometer in single-collector mode using a Faraday or Daly detector. A fractionation correction of 0.10% per amu for both Pb and U, and a Daly conversion factor of 0.4% per amu were applied. Decay constants used ($^{235}U = 9.8485 \times 10^{-10}$ y^{-1} and $^{238}U = 1.55125 \times 10^{-10}$ y^{-1}) are

from Jaffey et al. [57]. Uncertainties in isotopic ratios were estimated by propagating errors from internal ratio reproducibility, uncertainties in the fractionation correction (0.038% and 0.026% for Pb and U, respectively) determined from the long-term reproducibility of the NBS-SRM 981 common Pb and NBS-SRM U 500 standards, and the isotopic composition and amount of both laboratory blank and initial common Pb. Age regressions were calculated using the program of Davis [58].

6. U–Pb isotope results

The U–Pb isotopic results are summarized in Table 1 and shown graphically in Fig. 5. All errors given in the text and in Table 1, as well as those represented by ellipses in Fig. 5, are 2σ errors.

6.1. Inlandsee pseudotachylitic breccia (INL)

The zircon population from the Inlandsee pseudotachylitic breccia (sample INL) is composed mainly

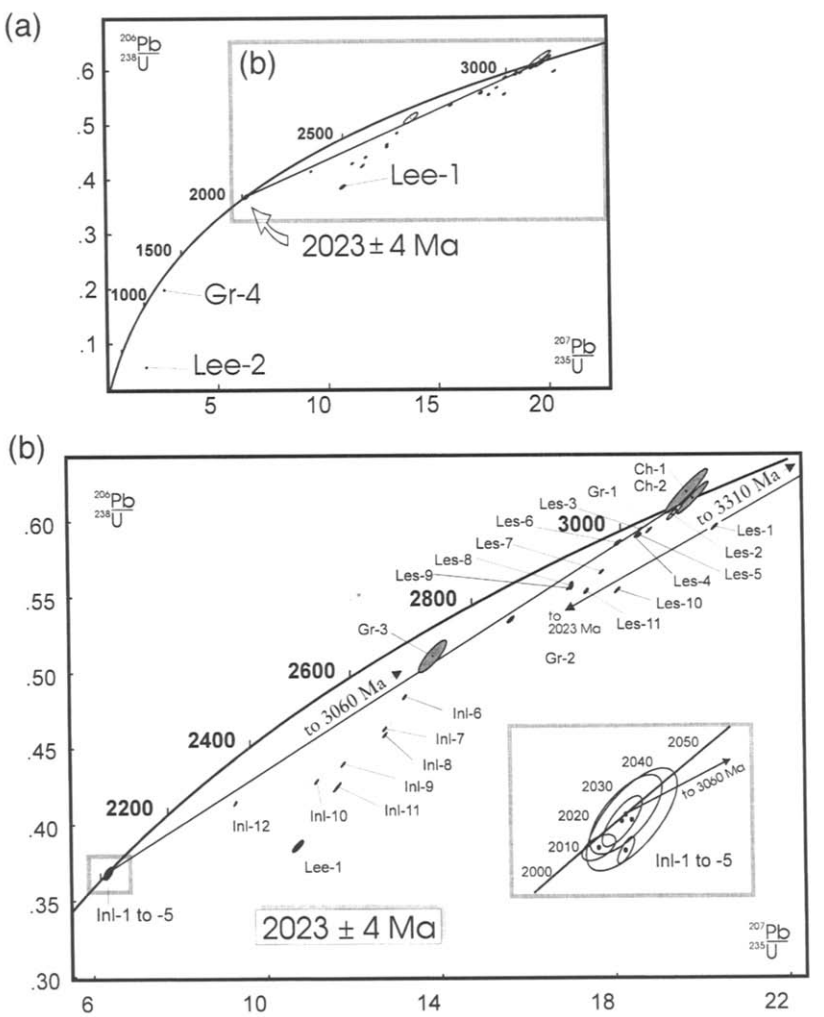


Fig. 5. Concordia diagrams showing U–Pb isotopic data for Vredefort zircon. (a) shows most of the concordia curve and includes all data from this study, in particular, Gr-4 and LEE-2. The lower right inset shows an enlargement of the data obtained for the 2023 Ma age for newly crystallized zircons from the Inlandsee pseudotachylitic breccia.

Table 1
U–Pb isotopic data for zircon from pseudotachylitic breccia and Granophyre, Vredefort Structure, South Africa

Sample	Fraction analysed	Weight (mg)	Pb (ppm)	U (ppm)	PbC (pg)	Isotopic ratios					Age (Ma)
						$^{206}Pb/^{204}Pb$	$^{208}Pb/^{206}Pb$	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	$^{207}Pb/^{206}Pb$
Inlandsee pseudotachylitic breccia											
INL-1	1, clear, pink, ab, unshocked	0.001	104	121	2	687	1.5704	0.3688 ± 34	6.338 ± 86	0.12466 ± 120	2024 ± 17
INL-2	1, clear, euhedral, pink, ab, unshocked	0.001	100	107	2	1602	1.8084	0.3682 ± 20	6.330 ± 46	0.12468 ± 58	2024 ± 8
INL-3	1, clear, pink, ab, unshocked	0.001	44	50	2	498	1.6350	0.3684 ± 42	6.352 ± 106	0.12506 ± 150	2030 ± 21
INL-4	1, clear, pink, ab, unshocked	0.002	23	34	1	1343	1.0414	0.3660 ± 12	6.274 ± 40	0.12432 ± 66	2019 ± 10
INL-5	3 fragments, clear, pink, not ab, unshocked	0.008	36	63	3	3920	0.6682	0.3658 ± 12	6.338 ± 24	0.12565 ± 30	2038 ± 4
INL-6	1, brown, cracks, PMs 1 direction	0.006	345	610	6	17089	0.1573	0.4836 ± 16	13.198 ± 46	0.19793 ± 28	2809 ± 2
INL-7	1, opaque, round, PMs 3 directions	0.005	221	399	15	3921	0.1744	0.4625 ± 13	12.731 ± 42	0.19962 ± 26	2823 ± 2
INL-8	1, cracks, partly opaque, round, fused PMs	0.005	243	439	7	9600	0.2598	0.4585 ± 18	12.734 ± 50	0.20142 ± 32	2838 ± 3
INL-9	1, round, cracks, fused PMs	0.004	242	488	17	3084	0.1004	0.4394 ± 19	11.763 ± 48	0.19417 ± 20	2778 ± 2
INL-10	1, brown, round, cracks, translucent, fused PMs	0.008	848	1527	416	794	0.2414	0.4244 ± 30	11.621 ± 84	0.19860 ± 32	2815 ± 3
INL-11	1, round, cracks, clear, fused PMs	0.005	87	143	5	3492	0.5195	0.4139 ± 14	9.280 ± 34	0.16259 ± 24	2483 ± 3
INL-12	1 fragment, opaque, cracks, fused PMs	0.003	237	487	18	2219	0.1084	0.4281 ± 16	11.145 ± 42	0.18880 ± 22	2732 ± 2
Lesutoskraal pseudotachylitic breccia											
LES-1	1, opaque, subround, PMs	0.005	221	307	4	14673	0.1549	0.5954 ± 22	20.356 ± 74	0.24794 ± 26	3172 ± 2
LES-2	1, clear, colorless tip, ab, no PMs	0.002	20	25	1	2814	0.2580	0.6034 ± 38	19.386 ± 120	0.23302 ± 60	3073 ± 4
LES-3	1, clear tip, ab, PMs	0.003	101	130	1	16812	0.2859	0.5937 ± 15	18.652 ± 52	0.22788 ± 30	3037 ± 2
LES-4	1, subround, clear, PMs 1 direction	0.003	75	98	3	3966	0.2693	0.5902 ± 20	18.574 ± 64	0.22822 ± 32	3040 ± 2
LES-5	1, subhedral, clear to translucent, PMs 3 directions	0.006	216	283	3	20288	0.2725	0.5899 ± 16	18.626 ± 52	0.22900 ± 26	3045 ± 2
LES-6	1, clear, pink euhedral, ab, no PMs	0.003	38	50	3	1718	0.2725	0.5848 ± 22	18.132 ± 80	0.22487 ± 54	3016 ± 4
LES-7	1, subhedral, translucent to opaque, PMs 3 directions	0.008	270	382	5	21864	0.2254	0.5659 ± 16	17.754 ± 54	0.22754 ± 22	3035 ± 2
LES-8	1, clear, coarse cracks, PMs uncertain	0.002	223	321	3	7949	0.2231	0.5567 ± 26	17.056 ± 48	0.22219 ± 90	2997 ± 6
LES-9	1, round, coarse parallel cracks 1 direction (possible PMs)	0.005	330	464	10	7737	0.2572	0.5551 ± 14	16.995 ± 46	0.22205 ± 20	2996 ± 1
LES-10	1, euhedral, PMs 3 directions	0.005	204	311	4	14385	0.1336	0.5537 ± 20	18.108 ± 68	0.23720 ± 30	3101 ± 2
LES-11	1, round, cracks, translucent, PMs uncertain	0.002	496	778	5	11279	0.1067	0.5532 ± 18	17.397 ± 54	0.22809 ± 40	3039 ± 3
Leeukop pseudotachylitic breccia											
LEE-1	1, euhedral, equant, short prismatic, ab, PMs uncertain	0.001	25	57	2	591	0.1027	0.3857 ± 39	10.702 ± 126	0.20122 ± 124	2836 ± 10
LEE-2	1 fragment, round, cracks, pink, polished, ab, PMs uncertain	0.001	45	666	3	973	0.1233	0.0574 ± 2	1.764 ± 10	0.22268 ± 74	3000 ± 5
Granophyre											
Gr-1	1, cracked, colorless, ab, no PMs	0.010	76	105	2	15756	0.1806	0.5931 ± 20	18.847 ± 66	0.23045 ± 30	3055 ± 2
Gr-2	1, cracked, colorless, ab, no PMs	0.002	45	75	2	2560	0.0979	0.5345 ± 24	15.658 ± 80	0.21245 ± 56	2924 ± 4
Gr-3	1, colorless, slightly hazy, ab, PMs uncertain	0.003	9	16	4	417	0.1183	0.5106 ± 106	13.854 ± 324	0.19679 ± 230	2800 ± 19
Gr-4	1, round, opaque white, granular	0.002	652	3148	62	1277	0.0692	0.1982 ± 6	2.592 ± 8	0.09484 ± 18	1525 ± 4
Charnockite											
Ch-1	1, subhedral, clear, ab, no PMs	0.001	18	24	2	399	0.1722	0.6187 ± 144	19.724 ± 488	0.23120 ± 172	3060 ± 12
Ch-2	1, subround, clear, ab, no PMs	0.001	24	31	2	746	0.2342	0.6143 ± 106	19.863 ± 360	0.23449 ± 112	3083 ± 8

ab = abraded [65]; PMs = planar microstructures (see text); PbC = the total common Pb from sample and laboratory sources. $^{206}Pb/^{204}Pb$ ratios have been corrected for spike and fractionation, all other isotopic ratios have been corrected for spike, fractionation, blank and initial common Pb. Errors are 2σ.

of brown grains, which are generally about 100–250 μm long (up to 400 μm long grains were observed), severely cracked, opaque, and rather rounded. Evidence for shock deformation in grains from this population occurred in the form of planar microstructures. In many of the grains examined it was unclear whether planar microstructures were present because the surfaces displayed discontinuous linear features and open cracks that appear to be fused, perhaps from superimposed effects resulting from partial annealing of the original shock features, due to high temperature post-impact heating (Fig. 4d). Similar observations were reported by Ostermann et al. [53] in zircons from an impact-generated dike from the Sudbury Structure.

A second population of clear, pale pink, about 100–150 μm long, gem-quality zircon grains was recovered from this sample (Fig. 4e). These crystals are euhedral and display perfect pyramidal faces, without any evidence of shock deformation. It is therefore assumed that this zircon population was formed during crystallization of the breccia matrix. Th/U ratios of 5 in these grains are relatively high in comparison to values of 0.3–1.0 for other grains in this study that show the least amount of discordance.

The U–Pb isotopic data for grains from the brown zircon population of sample INL (data points INL-6 to INL-12, Fig. 5b) indicate extensive Pb loss. The points plot in an array in the lower half of the concordia diagram and probably indicate Pb loss induced by impact and post-impact related metamorphism and subsequent recent Pb loss. Points INL-6 and INL-12 show the least amount of late Pb loss and appear to be partially reset close to a ca. 2000 Ma lower intercept.

Four gem-quality zircon grains from the second population (INL-1 to INL-4) in sample INL were each abraded prior to analysis. These grains gave concordant results with $^{207}Pb/^{206}Pb$ ages of 2024, 2024, 2030 and 2019 Ma, respectively, which overlap within their analytical uncertainties (see Fig. 5b, inset). The weighted mean age given by these four points is 2023 ± 4 Ma (2σ). A fifth zircon fraction, consisting of three fragments of gem-type zircon that were not abraded (INL-5), is 1.6% discordant and plots within error of two of the four concordant points, but was not included in the mean age calculation because of its slight discordance. Because of the 'igneous' appearance of this unshocked zircon population, that is clearly distinguished from the pre-impact zircon population, in which the grains exhibit some degree of rounding, probably due to an earlier Archean metamorphism, the age of 2023 ± 4 Ma is interpreted as the time of formation of the pseudotachylitic breccia. Accordingly, this age must be a very close approximation to the time of impact.

6.2. Lesutaskraal pseudotachylitic breccia (LES)

A single zircon population was recovered from pseudotachylitic breccia LES from the Lesutaskraal Quarry. These grains are brown, severely cracked, altered, translucent to opaque, and vary from euhedral to well rounded after resorption (Fig. 4f). The average grain sizes (in the longest dimension) vary from > 500 μm to < 120 μm. Most of these grains are shocked, as demonstrated by multiple sets of planar microstructures (Fig. 4a).

Eleven single zircon grains (LES-1 to LES-11) from pseudotachylitic breccia sample LES plot in a discordant array in the upper right quadrant of Fig. 5. The $^{207}Pb/^{206}Pb$ ages of the least discordant points (LES-1 to LES-6, which are 6.3%, 1.2%, 1.4%, 2.0%, 2.3%, and 2.0% discordant) are 3172, 3073, 3037, 3040, 3045, and 3116 Ma, respectively. These points give upper intercept ages between about 3060 and 3310 Ma, when constrained by the lower intercept age of 2023 Ma. This provides the best approximation of the primary ages for these shocked zircons and for the formation (or metamorphism, see Section 7) of the host target rocks. LES-3 and LES-6 indicate the least amount of post ca. 2023 Ma Pb loss and are co-linear on a line from ca. 3065 Ma to 2023 Ma.

6.3. Charnockitic gneiss (Ch)

The zircon population from the charnockitic gneiss from the Lesutaskraal Quarry consists of brown, cracked, euhedral to rounded, equant to prismatic (typically with 3:1 aspect ratios) crystals that are about 100–350 μm long (Fig. 4g). Shocked grains exhibit multiple sets of planar microstructures (Fig. 4h). A distinct subpopulation of small (ca. 50 μm long), unshocked, clear, crack-free, rounded to subrounded crystals was also recovered (Fig. 4g).

Two zircons from the unshocked variety of the charnockitic gneiss sample (Ch-1 and Ch-2) were analysed and gave concordant results with $^{207}Pb/^{206}Pb$ ages of 3060 ± 12 and 3083 ± 8 Ma, respectively. These ages, in comparison with the other data in the upper part of the concordia diagram, either reflect a minimum age for the formation of the basement granitoids, or a metamorphic overprint at about that time.

6.4. Leeukop pseudotachylitic breccia (LEE)

Zircons from the breccia sample from Leeukop Quarry (LEE) are opaque to translucent, 150–300 μm long, and have the shapes of partially resorbed prisms with 3:1 to 4:1 aspect ratios. Shock features in the form of planar microstructures are frequently visible under a binocular microscope, even without enhancement by etching. A very small population of smaller (50–110 μm), clearer, euhedral to subhedral zircons of equant shapes was recovered as well. Two of the clearest, least cracked grains from this type were selected for isotopic analysis (LEE-1 and LEE-2).

These gave extremely discordant results. LEE-1 (30% discordant) plots below the array of data defined by the Inlandsee borehole data, with a $^{207}Pb/^{206}Pb$ age of 2836 ± 30 Ma, and LEE-2 (90% discordant; Fig. 5a) has a $^{207}Pb/^{206}Pb$ age of 3000 ± 5 Ma. The preferred interpretation for this discordance is extreme recent Pb loss. Consequently, these data do not provide exact information regarding the age of the primary zircon source, nor about secondary metamorphic events or the age of the impact event.

6.5. Granophyre (Gr)

The Granophyre sample provided zircons with both multiple sets of planar microstructures (Fig. 4i) and granular texture (Fig. 4b,c). These crystals range in length from about 70 to 375 μm, are translucent to opaque (milky white), generally cracked and well rounded by resorption, with the smaller grains showing only a moderate degree of resorption. It was hoped in vain that this sample, collected mainly from the granular Granophyre variety, might yield clear, euhedral crystals, perhaps grown during formation of the Granophyre. Only a few apparently unshocked, small (100–250 μm) zircons, of variable characteristics (colorless, yellow and pink; euhedral to rounded; equant to prismatic; clear to hazy) were recovered.

Three grains from the subpopulation of small zircons from the Granophyre sample (Gr-1 to Gr-3) were analysed. They gave $^{207}Pb/^{206}Pb$ ages of 3055, 2924 and 2800 Ma and are 2.2%, 6.9% and 6.1% discordant, respectively. Gr-1 and Gr-2 are co-linear with the 2023 Ma lower intercept (11% probability of fit) and an upper intercept of 3106 ± 7 Ma. Gr-3 is within error of this line and may be derived from the same source as Gr-1 and Gr-2, or from a slightly younger source. Data for a shocked, opaque, granular zircon (Gr-4, Fig. 5a) indicate extreme discordance and may reflect a combination of complete to near-complete Pb loss, as a result of shock metamorphism at 2023 Ma, and superimposed younger Pb loss during a much younger event. The degree of surface-related younger Pb loss by diffusion would be enhanced by the extended surface area of the granular grain.

7. Summary and discussion

7.1. Primary zircon source ages

Shocked as well as some unshocked, but partially rounded or otherwise deformed, concordant, or nearly concordant zircons, from the Lesutaskraal breccia, charnockitic gneiss and from the Granophyre sample, indicate a range of upper intercept ages between about 3040, a minimum estimate, and 3300 Ma. Previous radiometric dating, as reviewed in this paper, only suggested a maximum age of 3.08 Ga for the Vredefort basement, although recent U–Pb zircon data indicate an age of 3292 ± 28 Ma [59] for a felsic gneiss from the Inlandsee leucogranofels.

The fact that several zircon ages for metamorphically rounded grains closely approach 3.08 Ga, and that a range of single zircon ages between 3.06 and 3.085 Ga was obtained, especially for Lesutaskraal zircons, leads us to propose that around that time a strong first metamorphic overprint occurred. This suggestion is in agreement with results from ^{40}Ar–^{39}Ar step heating studies [28,45] in which several

amphibole separates from charnockitic gneiss, from Lesutaskraal, and from a banded iron formation sample, from the Hospital Hill Subgroup of the collar of the Vredefort Dome, yielded plateau ages of 3.03 and 3.1 Ga. A 3.08 Ga event was previously recognized in the U–Pb and Rb–Sr whole rock dating results by Hart et al. [37]. The age of volcanic rocks in the upper part of the Dominion Group was determined to be 3074 ± 6 Ma (2σ) [60]. Clearly, a significant tectonomagmatic event occurred at that time in the region of the central Kaapvaal Craton.

7.2. Shock metamorphic textures and U–Pb discordance in zircon

Numerous zircon grains from all analysed Vredefort samples provided evidence for shock metamorphic deformation. Zircon crystals with one or multiple sets of planar microstructures were observed, and grains from the Granophyre sample also displayed the granular shock texture. These textures are well documented in impact-related rocks and have never been observed in terrestrial rocks that were deformed by processes other than impact metamorphism. These findings of shock metamorphosed zircons in Vredefort rocks, therefore, corroborate previous evidence for the impact origin of the Vredefort Structure.

It is interesting to note that, in contrast to the basal deformation features in quartz from Vredefort rocks, planar microstructures in zircon from our samples, with the exception of those from the INL sample, have not been strongly affected by the post-impact annealing event. This observation leads to the conclusion that shock metamorphic textures in zircon have better potential than the established shock effects in quartz to confirm the occurrence of impact structures, especially of old, deeply eroded and tectonically modified impact structures. Only the INL sample from the core of the structure contains zircons with apparently partially healed planar microstructures coexisting in the same breccia with newly crystallized zircons.

Although laboratory experiments have shown that shock wave metamorphism alone will not induce significant Pb loss [52], several studies [46–48,50] have demonstrated a correlation between the degree of discordance of U–Pb data with the intensity of shock-induced textural change in zircon crystals from the Sudbury and Chicxulub Structures, as well as corresponding ejecta from the latter. These data indicate that grains with minimal structural damage (i.e., that are unshocked or exhibit planar microdeformations) strongly retain their primary ages and may indicate minor Pb loss at the time of impact, whereas data for partly to completely granular grains show extensive Pb loss at the time of impact. Thus, it would appear from these examples that Pb loss in zircon is predominantly controlled by the intensity of impact-induced shock metamorphism. The fact that this correlation is observed in distal K/T boundary ejecta indicates that Pb loss occurred within seconds following impact and, given that a similar correlation is also seen in zircons from samples from within a crater structure [46,50], this evidence does not support the suggestion that Pb loss was induced by post-shock waste heat from local melt lithologies [61]. Zircons from Vredefort impact-generated lithologies may have lost Pb in a similar correlated fashion, but recent Pb loss in many of our grains complicates a clear correlation. Co-linear data for zircons from the Granophyre, however, suggest a common source age of ca. 3.1 Ga and Pb loss at ca. 2.0 Ga. The granular grain from this sample probably lost most of its Pb during shock metamorphism, and then during a subsequent later event at about ca. 1.0 Ga. Concordant results from the present study suggest that post-shock annealing at high temperatures cannot account entirely for the observed discordance patterns, although the annealed shock features and extreme Pb loss observed in zircons from the Inlandsee pseudotachylitic breccia from the core of the structure suggest that post-impact metamorphism affected this sample significantly.

7.3. The age of the Vredefort impact event

The suite of small, euhedral, clear, and unshocked zircon crystals separated from the Inlandsee pseudotachylitic breccia are part of a very fine grained, high-temperature mineral assemblage, that crystallized at temperatures above 700°C and that does not exhibit evidence for post-crystallization re-equilibration [62]. This observation is supported by petrological evidence from a large number of granulite samples from the Inlandsee Leucogranofels terrane, which all indicate that cooling below about 700°C

was very rapid. Evidence for rapid cooling is also provided by sample INL itself: the large, only marginally resorbed biotite grains in otherwise fully annealed clasts within this breccia do not show any evidence for thermal alteration/degradation. This is clear evidence that, after an initial high temperature (melting) event, cooling proceeded at a relatively rapid pace.

Concordant data for unshocked zircons give an age of 2023 ± 4 Ma and, because the zircons have not been subjected to alteration/deformation and isotopic disturbance, we interpret this age of 2023 Ma as the best currently available age for the impact event. This conclusion is, of course, based on the assumption that the pseudotachylitic breccia was formed during the impact event and not either before or after it. Based on the previous $^{40}Ar-^{39}Ar$ step heating and, especially, laser Ar work, in conjunction with detailed petrographic analysis [29,26,45], it is justified to assume that most of the pseudotachylitic breccias from the Vredefort Structure were formed in one event at about 2–2.02 Ga ago.

It can therefore be concluded that the 2023 ± 4 Ma age pinpoints the time of the Vredefort catastrophe. This represents a major refinement of the previous U–Pb zircon age of 2002 ± 52 Ma (unpublished data listed in Walraven et al. [22]). Our age also supports the conclusions drawn from the U–Pb dating by Moser [59], and argon dating by Spray et al. [29], Trieloff et al. [26], and Reimold et al. [45], that it is possible to separate the respective ages of the Vredefort event and Bushveld magmatism (at 2055 ± 2 Ma; Armstrong and Kamo, unpublished concordant U–Pb zircon data) by about 30 m.y. This distinction is of major importance for the understanding of regional metamorphic and metasomatic processes, which are of considerable economic importance with regard to the gold mineralization of the Witwatersrand basin [64].

7.4. Post-impact activity

Since the laser Ar dating of Spray et al. [29], it has been shown that a hydrothermal alteration event, related to tectonic and magmatic processes, affected pseudotachylitic breccias and other rocks from the Vredefort Structure in post-2000 Ma times, at ca. 1.0 Ga. Additional evidence for an important event between about 1000 and 1300 Ma (the Kibaran/Grenville event) was provided by $^{40}Ar-^{39}Ar$ step heating results on pseudotachylitic and cataclastic breccias from the Vredefort Dome and the western part of the Witwatersrand Basin, as well as of different minerals from Vredefort country rocks, by Reimold et al. [45]. Fission xenon dating on Witwatersrand uraninites [63] also yielded 0.85–1.35 Ga ages.

In our study, the granular zircon from the Granophyre shows evidence for Pb loss during this later event. Granular zircons lost most or all of their Pb during the impact-related recrystallization process [46–48,50]. Thus, this grain experienced two episodes of major Pb loss, first during the impact event and later during the Kibaran/Grenville event. This result supports the effectiveness of this event, which had previously only been recognized in thermochronological studies, but which obviously had the potential to reset this type of zircon grain as well. This event has recently been linked to the orogenic processes active between ca. 1.3 and 1.0 Ga at the southern margin of the Craton [63].

8. Conclusions

This study has provided further evidence that the Vredefort Structure was formed by the impact of a large extraterrestrial projectile. The observed shock metamorphic effects in zircon (planar microstructures and granular texture) from a number of different lithologies in the basement to the Vredefort Dome is unequivocal evidence in favor of impact.

The age of the impact event at 2023 ± 4 Ma is closely constrained by the U–Pb isotopic results for concordant zircons from the Inlandsee breccia sample. It has been discussed that textural evidence from pseudotachylitic breccias and Inlandsee Leucogranofels indicates a relatively short period of high-grade metamorphism post-dating the impact event. It is believed that this period did not extend beyond the time given by the analytical error on our impact age. To what extent this thermal event represents the waning regional, pre- to post-impact, Bushveld Complex related metamorphism, or is the result of directly impact-induced crustal heating, has not been resolved.

Zircon crystals from basement granitoid rocks of

the Vredefort Dome indicate ages between 3.1 and 3.3 Ga and are believed to represent the formation time for these granitoid rocks. In addition, it is possible that crystallization ages as young as 2.8 Ga are indicated by some discordant data.

Several, only slightly discordant, zircon grains yielded ages between 3.0 and 3.1 Ga, with a number of data around 3.08 Ga. In comparison with other published dating results, these data may indicate a metamorphic event, which probably coincides with the alteration event that affected the pyroxenes in the charnockitic gneiss.

Finally, Pb loss in zircon in Vredefort rocks occurred at 2.023 Ga, probably as the result of both shock metamorphism and post-impact thermal overprinting. In addition, a much later overprinting event was recognized, which appears to be related to Kibaran tectonomagmatic processes.

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